

Latent coverage of noisy calibration thresholds

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SUMMARY

Prediction intervals for a latent quantity, such as the mean of an unsampled area or a response measured with error, are often calibrated on noisy proxies. This is safe when the latent distribution and the noise are both symmetric and unimodal, and without a shape assumption it can double the error rate. We show that one shape condition on the latent distribution, bi-log-concavity, closes almost all of this gap, at a price computed exactly or within certified bounds for classes of noise laws and uniformly over the noise variances: noisy coverage 0.9 guarantees latent coverage 0.8991 under Gaussian noise, 0.8981 under any symmetric unimodal noise, heavy-tailed or not, and 0.8935 under any mean-zero log-concave noise. Neither half of the assumptions can be weakened to unimodality: for unimodal latent laws, even with mean zero, the guarantee falls to the shape-free value, 0.8 under median-zero noise, and under mean-zero unimodal noise nothing can be guaranteed. When the noise laws and variances differ between units and are unknown, we characterize the smallest rank at which the noisy conformal threshold covers the latent target with prescribed probability, exactly for symmetric unimodal noise and within certified bounds otherwise, a few ranks above the usual one; without the raise the latent reliability can tend to zero as the calibration sample grows. Under Gaussian noise the worst-case widening of the calibrated radius is of the order of the noise variance when the latent mean is away from the boundary of the interval, and of the order of the noise standard deviation only near it.

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Some key words: Conformal prediction; Log-concavity; Measurement error; Small area estimation; Tolerance interval.

1. INTRODUCTION

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Suppose that a residual W , the latent target minus a centre fitted on independent training data, is observed only through $V = W + e$, with noise e independent of W . A survey statistician who calibrates a prediction interval for the true mean of an unsampled area on the direct estimates of sampled areas is in this position (Fay & Herriot, 1979), as is a conformal analyst whose calibration responses carry measurement error (Einbinder et al., 2024; Cohen et al., 2025). The threshold t at which the noisy scores $|V|$ have coverage p is observable. We ask what latent coverage $\Pr(|W| \leq t)$ it guarantees, uniformly over a class of latent laws and of noise laws of every scale.

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Two answers are known, at the two extremes. If W and e are symmetric and unimodal, the theorem of Anderson (1955) gives $\Pr(|W + e| \leq t) \leq \Pr(|W| \leq t)$, so the noisy threshold is conservative (Einbinder et al., 2024). Without a shape assumption a median-zero noise turns a noisy level $1 - \alpha$ into a latent level $1 - 2\alpha$ (Guille-Escuret & Ndiaye, 2024), and this cannot be improved. LatentCP (Zheng et al., 2026) gives a marginal latent guarantee through a known forward model; Cohen et al. (2025) estimate the noise-free threshold by deconvolution, without a latent guarantee; conformal intervals for small areas (Bersson & Hoff, 2024) cover observable

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responses, conformal confidence intervals (Zhang & Tuoto, 2026) target the expectations of sampled units, and empirical Bayes and parametric small-area intervals (Armstrong et al., 2022; Ignatiadis & Wager, 2022; Hall & Maiti, 2006; Yoshimori & Lahiri, 2014) concern sampled units or rely on a model for the latent effects. A symmetric latent law is seldom plausible, and the factor two is costly.

We show that the gap between the two answers is closed, almost entirely, by bi-log-concavity of the latent law (Dümbgen et al., 2017), a condition that includes the log-concave laws and some multimodal ones but not symmetry, and we price what remains for each class of noise laws (§ 2, Table 1). The condition cannot be weakened to unimodality, and the noise condition cannot be weakened to mean-zero unimodality. For calibration samples whose noise laws and variances differ between units and are unknown, the price becomes a raise of the conformal rank by a few ranks, and we characterize the smallest valid rank, exactly for symmetric unimodal noise (§ 3); without it the latent reliability of the usual rank can tend to zero. For Gaussian noise of known variance, § 4 describes the worst-case radius: it is organized by the location of the latent mean, which decides whether the widening is of the order of the noise variance, as for a fixed smooth law (Jochmans & Weidner, 2024), or of the noise standard deviation.

2. COVERAGE TRANSFER

2.1. Classes of latent and noise laws

Let $\mathcal{L}$ be the class of log-concave laws on the real line, including point masses, and $\mathcal{B} \supset \mathcal{L}$ the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave. Both classes are closed under translation and scaling, and convolution with a log-concave law preserves them (Prékopa, 1973); the Supplementary Material treats $\mathcal{B}$. For $u > 0$ let E_u be exponential with rate u . Let $\mathcal{E}$ be a class of noise laws closed under scaling and negation, and let

$$\Psi_{\mathcal{E}}(q) = \sup_{\epsilon \in \mathcal{E}} E\{\min(1, qe^{-\epsilon})\}.$$

PROPOSITION 1. *Let ϵ have a law in $\mathcal{E}$ and be independent of Y , with $\log F_Y$ concave. If $\text{pr}(Y + \epsilon \leq 0) \geq p > \Psi_{\mathcal{E}}(q)$, then $F_Y(0) \geq q$. If $p < \Psi_{\mathcal{E}}(q)$, some exponential tail Y with $F_Y(0) < q$ and some ϵ with law in $\mathcal{E}$ have $\text{pr}(Y + \epsilon \leq 0) > p$.*

Proof. If $v = F_Y^{-1}(q) > 0$, the exponential tail $v + a/u - E_u$, $a = \log(1/q)$, u a supergradient of $\log F_Y$ at v , lies stochastically below Y , and adding ϵ preserves this, so $p \leq \text{pr}(a/u - E_u + \epsilon \leq 0) = E\{\min(1, qe^{-u\epsilon})\} \leq \Psi_{\mathcal{E}}(q)$, as $u\epsilon$ has a law in $\mathcal{E}$. If F_Y has an atom at v with $Y \geq v$, as for a point mass, then $p \leq \text{pr}(\epsilon < 0) \leq \lim_{u \rightarrow \infty} E\{\min(1, qe^{-u\epsilon})\} \leq \Psi_{\mathcal{E}}(q)$. Conversely, $a - E_1$ with noise ϵ attains $E\{\min(1, qe^{-\epsilon})\}$; shift it slightly to the right. $\square$

Exponential tails are extremal because concavity of $\log F_Y$ limits how fast mass can accumulate below the q -quantile. Applied at both ends of $[-t, t]$, to $W - t$ and to $-W - t$, Proposition 1 gives the two-sided transfer: let $p_{\mathcal{E}}(q)$ be the infimum of the noisy levels whose budget, however split between the ends, leaves latent budgets summing to at most $1 - q$, where an end with noisy budget α needs latent budget $1 - \lambda_{\mathcal{E}}(1 - \alpha)$ and $\Psi_{\mathcal{E}}\{\lambda_{\mathcal{E}}(p)\} = p$. Then noisy coverage above $p_{\mathcal{E}}(q)$ at threshold t implies latent coverage q for every $W \in \mathcal{B}$ and every noise with law in $\mathcal{E}$, and $p_{\mathcal{E}}(q) \geq \Psi_{\mathcal{E}}(q)$, because exponential tails at one end with small noise leave the other end untouched (Supplementary Material).

THEOREM 1. *Let $q > e^{-1}$. (i) For symmetric unimodal noise, $\Psi_{\mathcal{E}}(q) = (1 + qe^{-r})/2$, where $r > 0$ solves $(1 + r)e^{-r} = (1 + \log q)/q$, attained by uniform noise; $\Psi_{\mathcal{E}}$ is strictly convex, and*

Table 1. Latent coverage guaranteed by noisy coverage 0.9, the noisy level that guarantees latent coverage 0.9, and the smallest valid rank (Theorem 2) at $\delta = 0.05$

latent law	noise	latent coverage	noisy level	$k, K = 10^3$	$k, K = 10^4$
symmetric unimodal	symmetric unimodal	≥ 0.9	0.9	916	9050
$\mathcal{B}$	Gaussian	0.8991	(0.90079, 0.901]	917	9058–9060
$\mathcal{B}$	symmetric unimodal	0.8981	0.9017873...	918	9068
$\mathcal{B}$	mean-zero log-concave	0.8935	(0.90517, 0.906]	921–922	9101–9109
$\mathcal{B}$	mean-zero unimodal	none	1	–	–
unimodal, mean zero	median zero	0.8	0.95	962	9537
unimodal, mean zero	mean-zero log-concave	$1 - e/10$	$1 - 0.1/e$	974	9664

Coverages and levels are certified; a range of ranks reflects a level known within certified bounds, with the lower rank necessary and the upper sufficient. The first row is the usual high-probability rank, valid by Anderson’s theorem; the exact level for symmetric unimodal noise is $\Psi_{\mathcal{E}}(0.9)$ of Theorem 1(i), and the latent coverages are rounded down.

$p_{\mathcal{E}}(q) = \Psi_{\mathcal{E}}(q)$. (ii) For mean-zero log-concave noise, $\Psi_{\mathcal{E}}(q)$ is the supremum over centred log-affine laws on segments, and at $q = 0.9$ it lies in $[0.90517, 0.9052]$, the lower end given by the centred exponential law. (iii) For mean-zero unimodal noise, $\Psi_{\mathcal{E}}(q) = 1$.

Part (i) holds because symmetric unimodal laws are scale mixtures of uniform laws; convexity makes the latent budget a superadditive function of the noisy one, so the worst split puts the whole budget at one end. Part (ii) follows by localization with the constraint $E(\epsilon) \geq 0$ and a ball-arithmetic branch and bound, and part (iii) by a law with most of its mass spread far below $\log q$, balanced by a rare long excursion (Supplementary Material). For Gaussian noise, $\Psi_{\mathcal{E}}$ is computed in § 4.

PROPOSITION 2. Let $\beta_{\mathcal{E}} = \inf_{\epsilon \in \mathcal{E}} \text{pr}(\epsilon \geq 0) > 0$ and $1 - \beta_{\mathcal{E}} < p < 1$. Then $\text{pr}(|W| \leq 1) \geq 1 - (1 - p)/\beta_{\mathcal{E}}$ for every W and every noise with law in $\mathcal{E}$ and $\text{pr}(|W + e| \leq 1) \geq p$. If the infimum is approached by laws continuous at 0, then for every $\gamma > 0$ there are a unimodal W with $E(W) = 0$ and such a noise with $\text{pr}(|W| \leq 1) \leq 1 - (1 - p)/\beta_{\mathcal{E}} + \gamma$. If $\beta_{\mathcal{E}} = 0$, approached in the same way, then for every $p < 1$ and $\gamma > 0$ there are such W and noise with $\text{pr}(|W| \leq 1) \leq \gamma$.

So for unimodal latent laws the guarantee is the shape-free one: $2p - 1$ for median-zero noise such as Gaussian or symmetric unimodal noise, $1 - e(1 - p)$ for mean-zero log-concave noise, since then $\beta_{\mathcal{E}} = e^{-1}$, attained by the centred exponential law, and nothing for mean-zero unimodal noise, where $\beta_{\mathcal{E}} = 0$. The extremal laws place a thin layer of mass just outside the threshold, a fraction $1 - \beta_{\mathcal{E}}$ of which the noise carries inside; bi-log-concavity forbids such a layer, since concavity of $\log F$ makes f/F nonincreasing, so the density cannot jump up at the threshold.

3. CALIBRATION WITH UNKNOWN NOISE

Let $W_1, \dots, W_K, W_{\text{new}}$ be independent with law G , and let $e_i = \sigma_i \epsilon_i$, independent of them and of each other, where the laws of the ϵ_i lie in $\mathcal{E}$ and may differ between units and the scales $\sigma_i > 0$ are unknown. The scores $V_i = W_i + e_i$ are independent but not identically distributed, and $\mathcal{D}$ denotes the calibration data. Let $T = |V|_{(k)}$.

THEOREM 2. If $G \in \mathcal{B}$, then

$$\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq T \mid \mathcal{D}) \geq q\} \geq \text{pr}\{\text{Bin}(K, p_{\mathcal{E}}(q)) \leq k - 1\}.$$

Conversely, if $\text{pr}\{\text{Bin}(K, \Psi_{\mathcal{E}}(q)) \leq k - 1\} < 1 - \delta$, there are a log-concave G and a noise law in $\mathcal{E}$, the same in every unit, for which the left side is below $1 - \delta$. So the smallest rank valid

110 for every $G \in \mathcal{B}$ and all such noise is the smallest k whose δ -quantile of the $\text{Beta}(k, K + 1 - k)$ distribution is at least $p_{\mathcal{E}}(q)$, when $p_{\mathcal{E}}(q) = \Psi_{\mathcal{E}}(q)$, and lies between the ranks given by $\Psi_{\mathcal{E}}(q)$ and $p_{\mathcal{E}}(q)$ otherwise.

Proof. Let $r = Q_q(|W|)$ for $W \sim G$. Latent coverage at T is at least q if and only if $T \geq r$, that is, if and only if $N = \#\{i : |V_i| < r\} \leq k - 1$. For $s < r$, $\text{pr}(|W| \leq s) < q$, so by the transfer of § 2 at threshold s , $\text{pr}(|V_i| \leq s) \leq p$ for every $p > p_{\mathcal{E}}(q)$, whatever the law and scale of e_i ; letting $s \uparrow r$, $\text{pr}(|V_i| < r) \leq p_{\mathcal{E}}(q)$. N is a sum of independent indicators with these success probabilities, so it is stochastically smaller than $\text{Bin}\{K, p_{\mathcal{E}}(q)\}$. Conversely, for $p < \Psi_{\mathcal{E}}(q)$ the one-ended laws of § 2 give a log-concave G and a noise with $\text{pr}(|V_i| < r) \geq \text{pr}(|V_i| \leq 1) > p$ and $r > 1$, so the left side is at most $\text{pr}\{\text{Bin}(K, p) \leq k - 1\}$, which tends to $\text{pr}\{\text{Bin}(K, \Psi_{\mathcal{E}}(q)) \leq k - 1\}$ as $p \uparrow \Psi_{\mathcal{E}}(q)$.
 120 The last statement is the identity $\text{pr}\{\text{Bin}(K, p) \leq k - 1\} = \text{pr}\{\text{Beta}(k, K + 1 - k) \geq p\}$. $\square$

The comparison is with each unit's own noise law, so neither the laws nor the variances need be known or shared, and no continuity is needed. At $q = 0.9$ and $\delta = 0.05$ a valid rank exists for $K \geq 29$ under Gaussian or symmetric unimodal noise and $K \geq 31$ under mean-zero log-concave noise, against $K \geq 59$ for a unimodal latent law and median-zero noise. For $K = 110$ the bi-
 125 log-concave rows with Gaussian, symmetric unimodal or mean-zero log-concave noise all give $k = 105$, the usual high-probability rank, against 109 without a shape assumption; the marginal rank $\lceil 0.9(K + 1) \rceil = 100$ is not covered. As K grows the raise grows, because the slack becomes large relative to the spacing of the levels, and it is necessary:

PROPOSITION 3. Let $q \in (0, 1)$. There are a log-concave law G and $D > 0$ such that, with
 130 $e_i \sim N(0, D)$ for all i , $\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq |V|_{(k_K)} \mid \mathcal{D}) \geq q\} \rightarrow 0$ as $K \rightarrow \infty$ for every sequence of ranks with $k_K/K \rightarrow q$, such as the marginal rank and, for any fixed δ , the usual high-probability rank.

Proof. By Theorem 3 some log-concave W has $H(1) \geq q$ and $r_q = Q_q(|W|) > 1$, where H is the distribution function of $|W + D^{1/2}Z|$, $Z \sim N(0, 1)$. As in the proof of Theorem 2, the latent
 135 reliability of rank k is $\text{pr}\{\text{Bin}(K, H(r_q-)) \leq k - 1\}$, and $H(r_q-) > H(1) \geq q$ as H is strictly increasing, so it tends to zero by the law of large numbers. $\square$

The loss in coverage is small and the loss in reliability is not. At the stress-test law of the Supplementary Material, with $D^{1/2}$ equal to 1% of the noisy q -quantile, the latent coverage tends to 0.89918, yet the usual rank has latent reliability 0.943, 0.918 and 0.151 at $K = 10^3, 10^4$ and
 140 10^6 , against 0.954, 0.958 and 0.989 for the rank of Theorem 2. For $29 \leq K \leq 300$ the shortfall is at most 0.005, so the raise matters for conformal calibration on large sets of noisy regression responses more than for small areas.

4. GAUSSIAN NOISE AND THE LOCATION OF THE MEAN

4.1. The sharp radius

145 For Gaussian noise $e \sim N(0, D)$ with D known, a radius can replace the raise. After scaling t to 1, write $x = D/t^2$, let $Z \sim N(0, 1)$ be independent of W and define

$$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{L}, \text{pr}(|W + x^{1/2}Z| \leq 1) \geq p\}, \quad (1)$$

where Q_q is the lower q -quantile, and $R_{p,q}(x) = -\infty$ if nothing is feasible. If calibration certifies noisy coverage p at threshold t , then $t R_{p,q}(D/t^2)$ is the smallest radius that guarantees latent coverage q for every log-concave latent law; this is sharpness within the class and for one
 150 constraint, not optimality among procedures that use the whole calibration sample. Let $R_{p,q}^{\mathcal{B}} \geq$

$R_{p,q}$ be defined with $\mathcal{B}$ in place of $\mathcal{L}$, and $R_{p,q}^{(b)}$ with the additional constraint $|E(W)| \leq b$, so that $1 - b$ bounds the distance of the mean from the boundary. Write $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1)$ and let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $e^{\beta w}$ on a segment.

PROPOSITION 4. $R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F}, E\{g_x(W)\} \geq p\}$. The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of § 4.3. 155

The proof uses the localization theorem of Fradelizi & Guédon (2004); over $\mathcal{F}$ the problem is a two-dimensional maximization of closed forms. Near each end of the interval the problem is one-sided. For the exponential tail $b - E_u$ let

$$m_u(b) = \text{pr}(b - E_u + Z \leq 0) = \Phi(-b) + e^{-ub+u^2/2} \Phi(b - u),$$

let $b_u(p)$ solve $m_u(b) = p$, and write $v_q(u) = b_u(q) - u^{-1} \log(1/q)$ and $k_q(u) = u^{-1} - b_u(q)$; the law $b_u(q) - E_u$ has q -quantile $v_q(u)$ and mean $-k_q(u)$. 160

LEMMA 1. Let $c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, \text{pr}(Y + Z \leq 0) \geq p\}$ and $c_q = c_{q,q}$. Then $c_{p,q} = \sup_{u>0}\{b_u(p) - u^{-1} \log(1/q)\}$, $c_q > 0$, and $q \mapsto c_q$ is nonincreasing. The same holds with $\mathcal{B}$ in place of $\mathcal{L}$.

Ball arithmetic gives $c_{0.8} = 0.04607$, $c_{0.9} = 0.01906$ and $c_{0.95} = 0.00841$.

4.2. Three regimes 165

THEOREM 3. For every $q \in (0, 1)$, every $x > 0$ and every $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$, $Q_q(|W|) \leq 1 + c_q x^{1/2}$. Moreover $R_{q,q}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$, and likewise for $R_{q,q}^{\mathcal{B}}$.

Proof of the upper bound. Let $\alpha_R = \text{pr}(W + x^{1/2}Z > 1)$ and $\alpha_L = \text{pr}(W + x^{1/2}Z < -1)$, so that $\alpha_L + \alpha_R \leq 1 - q$. The law of $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave and $\text{pr}(Y + Z \leq 0) = 1 - \alpha_R$. By Lemma 1, $F_Y^{-1}(1 - \alpha_R) \leq c_{1-\alpha_R} \leq c_q$, that is, $\text{pr}(W > 1 + c_q x^{1/2}) \leq \alpha_R$. The same argument applied to $-W$, which uses concavity of $\log(1 - F_W)$, gives $\text{pr}(W < -1 - c_q x^{1/2}) \leq \alpha_L$, so $\text{pr}(|W| \leq 1 + c_q x^{1/2}) \geq q$. 170 $\square$

The lower bound uses boundary-layer laws $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$, whose right end point lies just outside the threshold. Since the constraint is feasible only if $x^{1/2} \leq 1/\Phi^{-1}\{(1 + q)/2\}$, the widening is below 1.16% of the radius at $q = 0.9$ for every $W \in \mathcal{B}$, and below 0.22% for log-concave laws (Supplementary Material). 175

THEOREM 4. Let $W \in \mathcal{B}$ with $|E(W)| \leq B$, let $\sigma > 0$ and $t > B$, and suppose that $\text{pr}(|W + \sigma Z| \leq t) \geq q$. Then

$$Q_q(|W|) \leq B + \{(t - B)^2 + \sigma^2\}^{1/2} \leq t + \frac{\sigma^2}{2(t - B)}.$$

In particular $R_{p,q}^{(b)}(x) \leq 1 + x/\{2(1 - b)\}$ for all $p \geq q$ and $x > 0$.

A mean far from an end of the interval caps how steeply the density can rise towards it; the proof makes this precise with the heat equation for quantiles (Steinbrecher & Shaw, 2008), which gives $d\{r(s) - B\}^2/ds \geq -1$ for $r(s) = Q_q(|W + s^{1/2}Z|)$. The order x cannot be improved, even for mean-zero laws: centred log-affine laws on a segment attain it (Supplementary Proposition S1), so centring does not replace symmetry. 180

THEOREM 5. Let $1 - e^{-1} < q < 1$, $x > 0$ and $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$. Then $|E(W)| < 1$, and if $\Delta = Q_q(|W|) - 1 > 0$,

$$2\{1 - |E(W)|\}\Delta + \Delta^2 \leq x. \quad (2)$$

Moreover, for every $\kappa \geq 0$, with $b_x = 1 - \kappa x^{1/2}$,

$$\lim_{x \rightarrow 0} x^{-1/2} \{R_{q,q}^{(b_x)}(x) - 1\} = L_q(\kappa) = \sup\{v_q(u) : u > 0, k_q(u) \geq \kappa\},$$

and the same limit holds with $\mathcal{L}$ replaced by $\mathcal{B}$ in the definition of $R^{(b)}$.

The function L_q equals c_q up to κ_q^* , about 20 at $q = 0.9$, and then decays like $\{1 - \log(1/q)\}/(2\kappa)$ (Supplementary Material). In the original units, the widening is of order $D^{1/2}$ only when the mean lies within a bounded number of noise standard deviations of the threshold, and of order D otherwise.

4.3. Slack and known variances

Splitting the noisy and latent budgets between the ends gives $Q_q(|W|) \leq 1 + C_{p,q}x^{1/2}$ for every x and every $W \in \mathcal{B}$ feasible at level $p \geq q$, where $C_{p,q} = \sup_{\alpha_L + \alpha_R = 1-p} \inf_{\beta_L + \beta_R = 1-q} \max(c_{1-\alpha_L, 1-\beta_L}, c_{1-\alpha_R, 1-\beta_R}) \geq c_{p,q}$.

COROLLARY 1. If $C_{p,q} \leq 0$ then $R_{p,q}^{\mathcal{B}}(x) \leq 1$ for every x , and if $c_{p,q} > 0$ then $R_{p,q}(x) > 1$ for all small x . At $q = 0.9$ the smallest slack $p - q$ with $C_{p,q} \leq 0$, and with $R_{p,q}(x) \leq 1$ for all x , lies in $(7.9 \times 10^{-4}, 10^{-3}]$. So for Gaussian noise $p_{\mathcal{E}}(0.9) \in (0.90079, 0.901]$, and noisy coverage 0.9 gives latent coverage 0.8991 for every $W \in \mathcal{B}$ and noise variance, while 0.8992 fails for some log-concave laws.

With known variances D_i the radius can be shortened. Let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and p_k the δ -quantile of the Beta($k, K + 1 - k$) distribution.

PROPOSITION 5. If $k \geq Kp_k + 1$ then $\text{pr}\{\bar{F}(T) \geq p_k\} \geq 1 - \delta$.

The proof uses Hoeffding (1956). On this event the average kernel $\bar{g}(w) = K^{-1} \sum_i \text{pr}(|w + e_i| \leq T)$ has $\int \bar{g} dG \geq p_k$, so by Proposition 4 the computed radius $s = TR^{\text{mix}}$, with R^{mix} defined by (1) with the constraint $\int \bar{g} dG \geq p_k - \omega$ and ω bounding a discretization error, covers the latent target with probability $1 - \delta$ for log-concave G ; our implementation returns a numerical upper bound on R^{mix} , not a certificate. If only a lower bound $D_{\min}$ is known, the shrunk threshold $\max\{0, T + C_{p',q}D_{\min}^{1/2}\}$ has the guarantee whenever $q \leq p' \leq p_k$ and $C_{p',q} < 0$; at $q = 0.9$, $K = 110$ and $\delta = 0.05$ this is $T - 0.114D_{\min}^{1/2}$ (Supplementary Material).

5. NUMERICAL ILLUSTRATION

The target is $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.9\} \geq 0.95$, with $K = 110$, $\text{var}(W) = 1$ and Gaussian noise with heterogeneous variances $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7. Latent residuals, all with mean zero, are drawn directly; conditional coverage given each data set is exact, and reliability is estimated from 150 data sets per law. Besides four standard log-concave laws we use a truncated exponential law in $\mathcal{F}$, a bimodal law in $\mathcal{B}$ but not in $\mathcal{L}$, and a t_3 law outside $\mathcal{B}$. We compare the noisy threshold at rank 105, the shrunk threshold and the computed radius of § 4.3, a Fay–Herriot tolerance interval adapted from Chatterjee et al. (2008), and the Markov rule, the radius r with $\bar{g}(r) = \{p_k - q\bar{g}(0)\}/(1 - q)$, the smallest the single constraint allows without a shape assumption, at its best rank, 107 (Supplementary Material).

Table 2. Reliability, times 100, and mean width (in parentheses); target 95

latent law	Markov	Fay–Herriot	noisy threshold	shrunk	computed
normal	100 (5.92)	96 (3.93)	100 (4.96)	100 (4.90)	98 (4.45)
Laplace	100 (6.29)	93 (3.96)	100 (5.15)	100 (5.09)	99 (4.66)
centred gamma	100 (6.16)	100 (3.95)	100 (4.97)	100 (4.90)	100 (4.45)
truncated Laplace	100 (5.89)	85 (3.89)	100 (4.93)	100 (4.87)	97 (4.42)
truncated exponential	100 (6.17)	95 (3.93)	100 (5.06)	100 (4.99)	100 (4.55)
bimodal ($\mathcal{B}$ only)	100 (5.77)	100 (4.01)	100 (4.87)	100 (4.81)	99 (4.35)
t_3 (outside $\mathcal{B}$)	100 (5.70)	97 (3.80)	100 (4.66)	100 (4.60)	98 (4.09)

Noisy threshold, T at rank 105; shrunk, $\max\{0, T - 0.114D_{\min}^{1/2}\}$ with $D_{\min} = \min_i D_i$; computed, TR^{mix} ; centred gamma with shape 2; truncated exponential, density proportional to e^{4w} on $[0, 1]$, standardized; bimodal, the mixture of Supplementary Lemma S1; widths are full widths. The Markov rule is guaranteed for every law; under the bimodal law the noisy and shrunk thresholds are guaranteed and the computed radius is exploratory; under t_3 the noisy threshold is guaranteed by Anderson’s theorem. Standard errors are at most 2.9 for reliabilities and 0.08 for widths.

The rules with a guarantee reached the target for every law in their class, and the computed radius did so also under the bimodal and t_3 laws, for which it has none. The computed radius was 9.5–12% shorter than the noisy threshold and 24–28% shorter than the Markov rule; The Fay–Herriot interval was 7–15% shorter than the computed radius but missed the target under the truncated Laplace law by 5.8 standard errors. With 2000 data sets per law the noisy and shrunk thresholds reached conditional coverage 0.9 in every data set, with mean conditional coverage 0.97–0.99: the guarantees are conservative for laws far from the extremal ones. With the same information, no rule being told the noise law or the variances, the ranks of Theorem 2 were compared under Gaussian, uniform, Laplace, t_3 and centred exponential noise and four latent laws (Supplementary Material). Every rule had reliability at least 0.998 wherever it is guaranteed. Against the usual rank and over the settings each rule covers, the thresholds for Gaussian, symmetric unimodal and mean-zero log-concave noise were at most 0.4%, 1.0% and 2.9% wider at $K = 1000$, and not wider at $K = 110$, while the shape-free rules were 20–54% wider at $K = 1000$: the latent shape condition reduces the cost of the guarantee by an order of magnitude. In the Supplementary Material LatentCP (Zheng et al., 2026) is 16–18% longer than the computed radius, and at oracle-matched reliability the gains lie in the cost of a guarantee, not in the shape of the interval.

6. DISCUSSION

The guarantees are conditional on the calibration sample and assume noise independent of the latent residual and a bi-log-concave, or for the computed radius log-concave, latent law shared by the calibration units and the new one. The target is an unsampled unit; for a sampled one the direct estimate is itself informative (Zhang & Tuoto, 2026). The shared law is the substantive assumption, and the noise is not where the risk lies: if the new residual is shifted by at most ρ latent standard deviations and the latent law is log-concave, the latent coverage can fall by at most ρ , and by $q(1 - e^{-\rho}) = q\rho + O(\rho^2)$ for some laws, so the linear order cannot be improved (Supplementary Material); Gaussian, symmetric unimodal or mean-zero log-concave noise costs at most 0.0065. Bi-log-concavity cannot be checked on the latent scale, but convolution with log-concave noise preserves it, so with equal variances a confidence band for the signed noisy residuals containing no bi-log-concave distribution function (Dümbgen et al., 2017) rules it out, though not conversely. Proposition 2 shows what is at stake: a latent law with a thin layer of mass

just beyond the threshold, which unimodality allows, costs the full shape-free factor. It shows that unimodality cannot replace bi-log-concavity, not that bi-log-concavity is necessary.

Open questions include an analytic proof that the centred exponential law is extremal among mean-zero log-concave noise laws, convexity of $\Psi_{\mathcal{E}}$ for Gaussian and log-concave noise, which would make their two-sided levels exact as in Theorem 1(i), and whether $R_{p,q}^{\mathcal{B}} = R_{p,q}$.

DECLARATION OF THE USE OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

[To be completed by the author.]

SUPPLEMENTARY MATERIAL

The Supplementary Material contains the omitted proofs, the certificates, further simulations and an application. Code reproducing every number is at <https://github.com/kunhailP/latent-coverage-noisy-thresholds>, tag biometrika-submission.

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